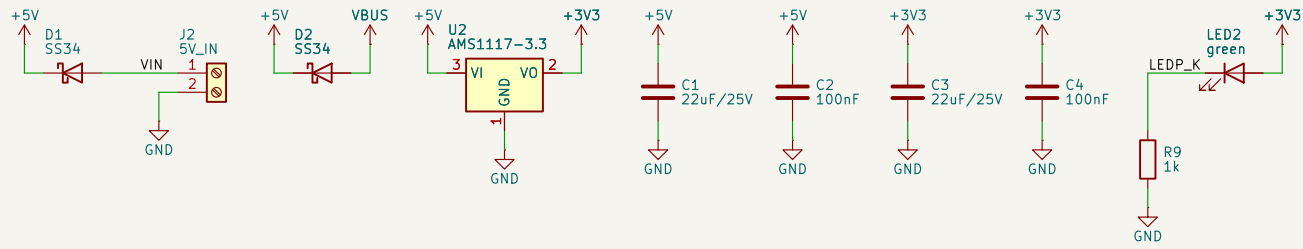
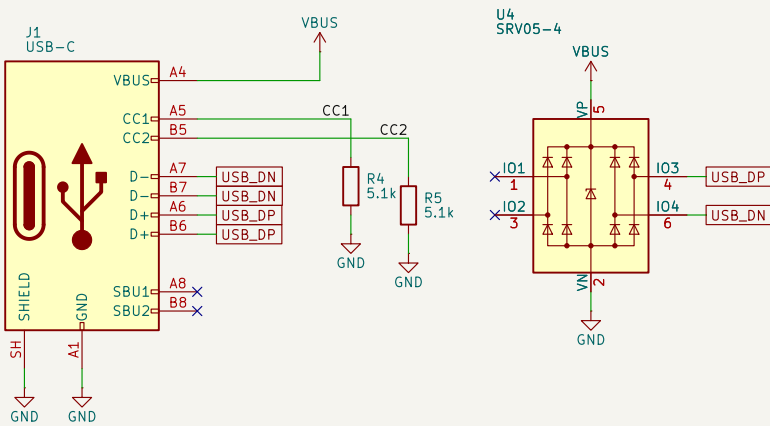


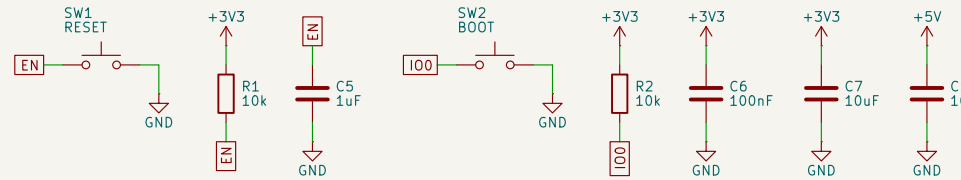
POWER IN
5V DC terminal or USB, Ored Schottky diodes



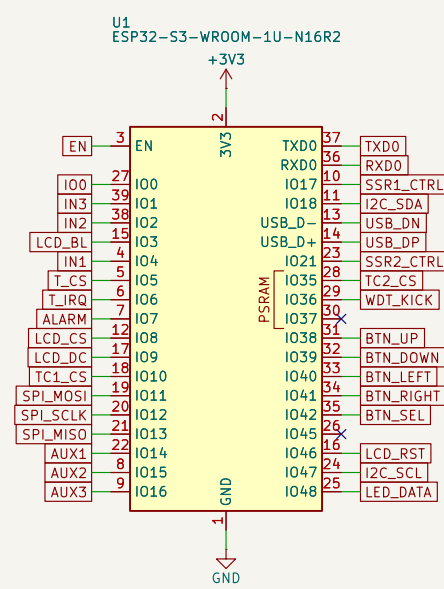
USB-C
native USB flashing + ESD



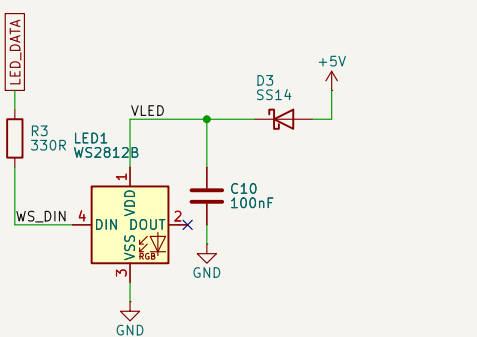
RESET / BOOT / DECOUPLING



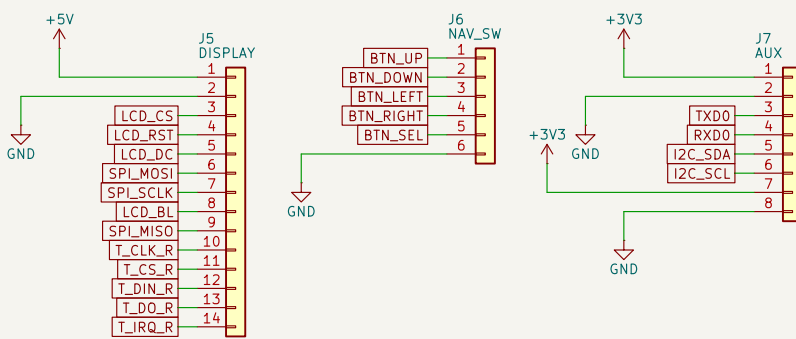
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



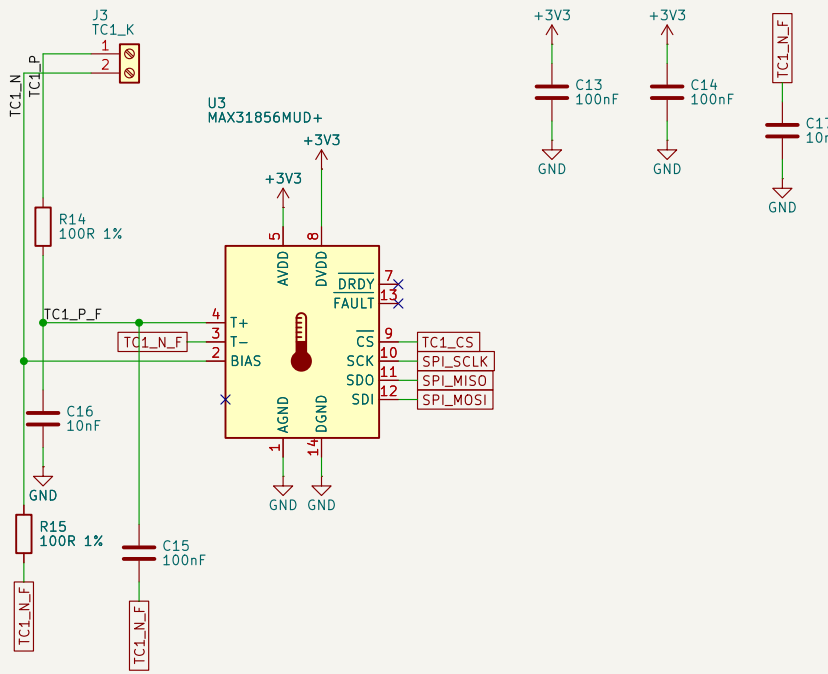
WS2812B STATUS LED
VDD dropped -4.6V for 3.3V data margin



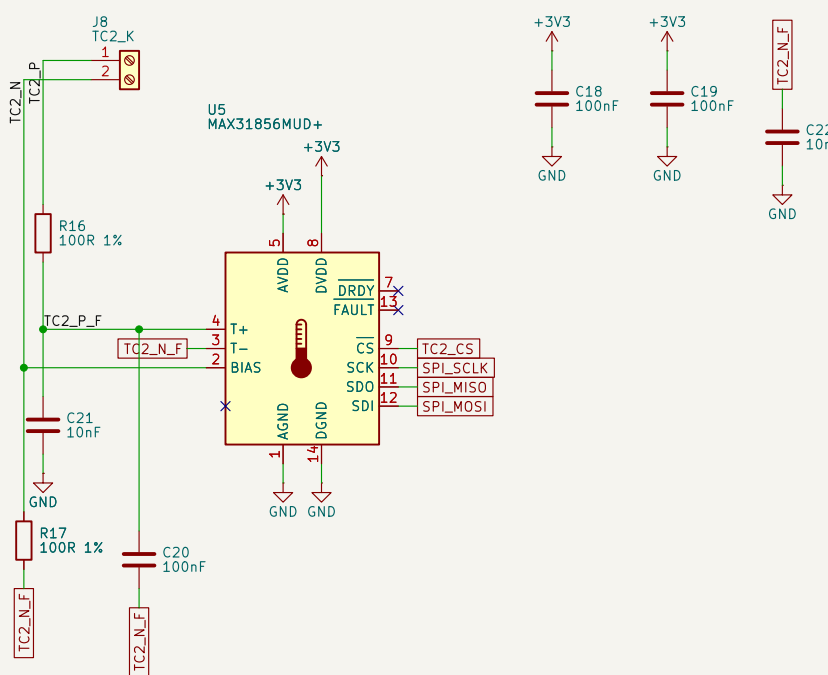
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



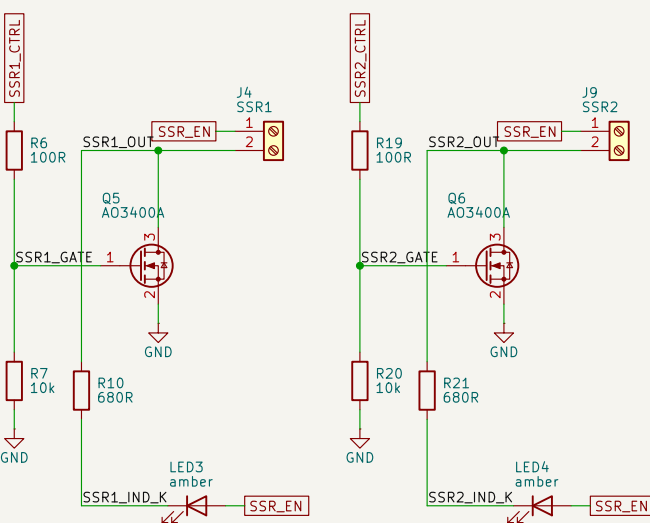
THERMOCOUPLE 1 (control) MAX31856
I- floats to J3, biased via BIAS



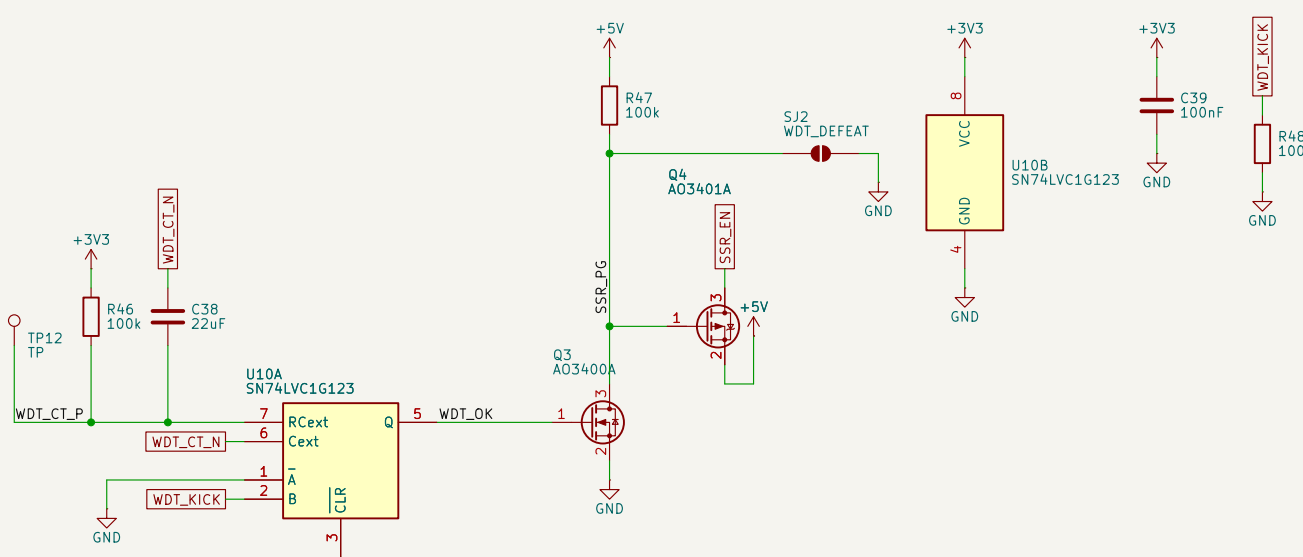
THERMOCOUPLE 2 (load) MAX31856
I- floats to J6, biased via BIAS



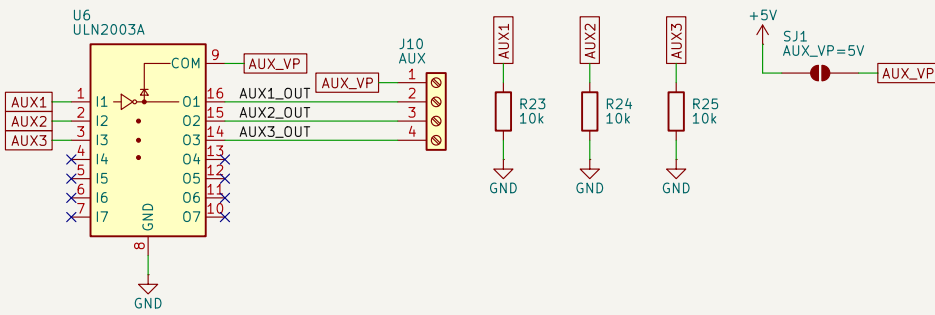
SSR DRIVE x2
low-side A03400A; indicator LED across each terminal



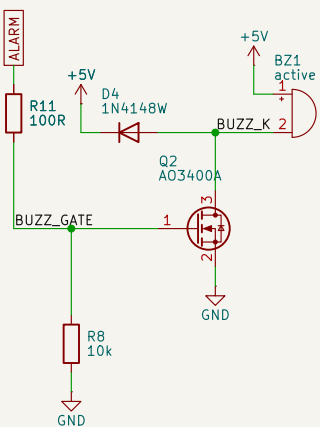
HARDWARE WATCHDOG
U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN - the +5V rail feeding BOTH SSR terminals and both indicators.
R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. S.J2 = bring-up defeat, REMOVE for service.



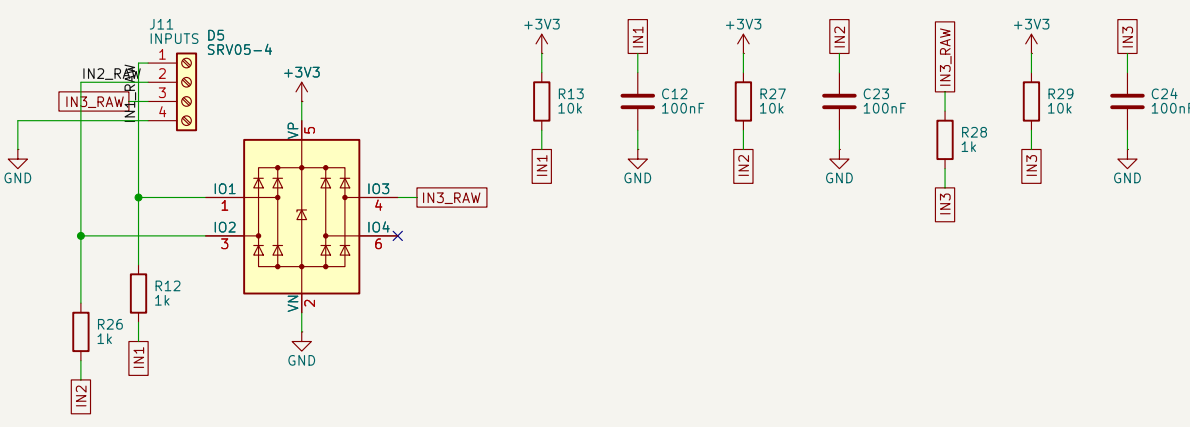
AUX OUTPUT BANK
ULN2003A vent/purge/spare; COM->AUX_VP, SJ1 links to +5V



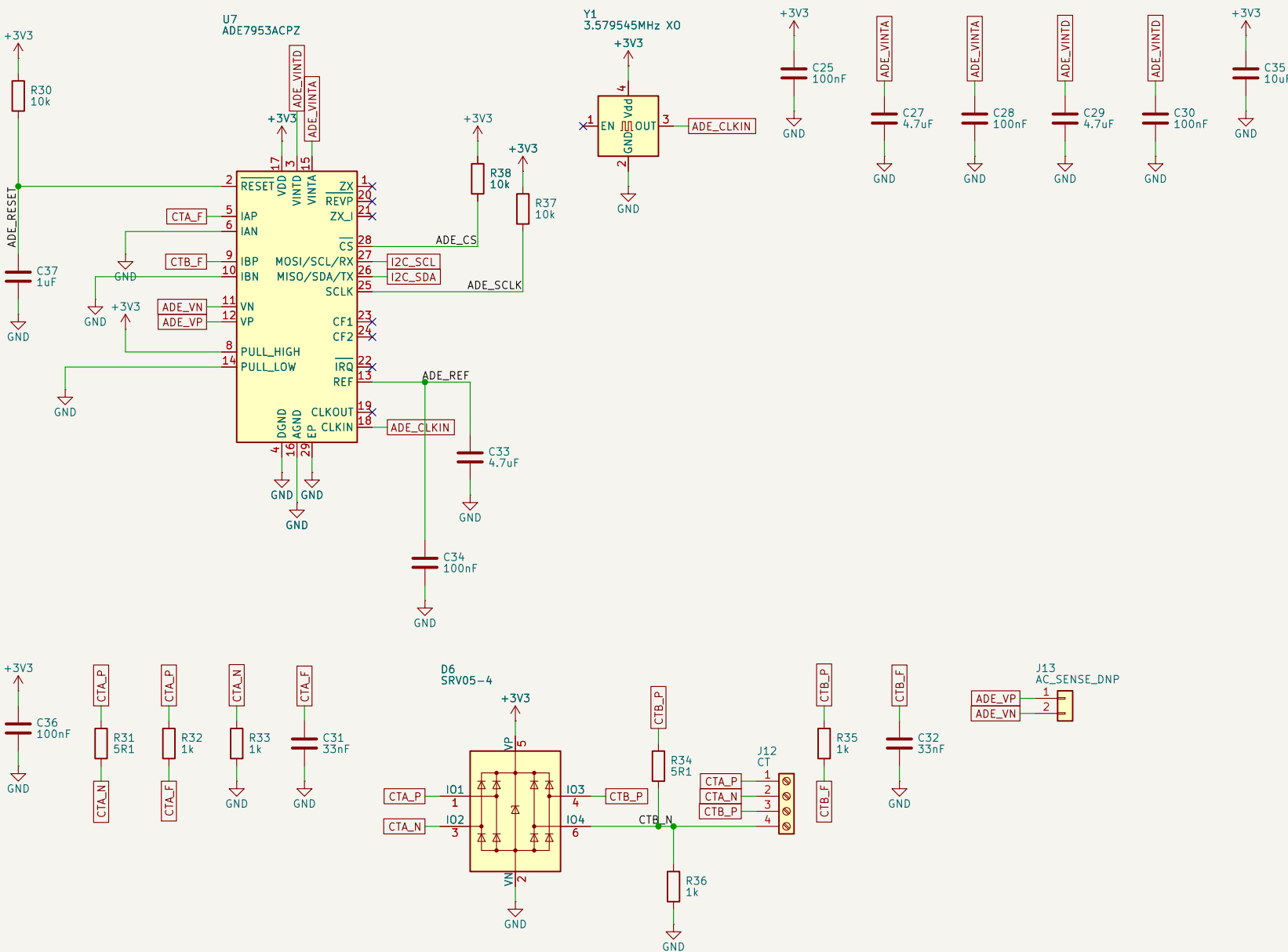
ALARM BUZZER



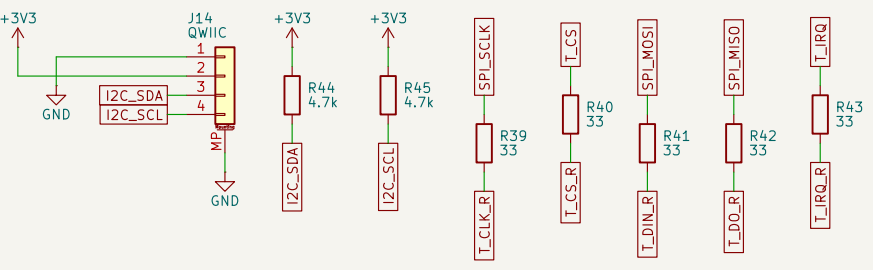
PROTECTED INPUTS x3 Iid / gas-flow / spare
1k + 10k pull-up + 100nF each, SRV05-4 TVS



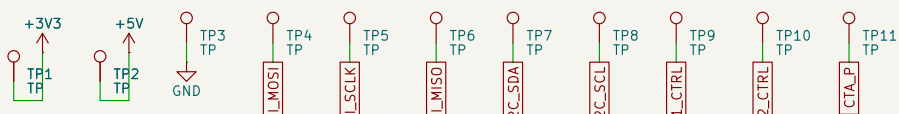
CT CURRENT SENSING ADE7953, I2C, current-only
no mains - VP/VN to DNP J13



TOUCH DAMPING + I2C EXPANSION
R39-R43 damp the shared SPI2 bus for the display module's XPT2046 (not on this board); J14 Qwicc and J7 5-8 (0.1 in) share the I2C bus, pulled up by R44/R45



TEST POINTS



MOUNTING / FIDUCIALS / POWER FLAGS
FID1-3 are bare-copper pick-and-place targets, on no net



NOTES - Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated).
pin2 = the switched low side, Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance.
NOT opto-isolated - rev B tried that and reversed it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.
TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required; two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted: inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (Iid/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST77965 + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=C5 4=RS1 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=I2C 11=I2C 12=I2C 13=I2C 14=I2C 15=I2C 16=I2C 17=I2C 18=I2C 19=I2C 20=I2C 21=I2C 22=I2C 23=I2C 24=I2C 25=I2C 26=I2C 27=I2C 28=I2C 29=I2C 30=I2C 31=I2C 32=I2C 33=I2C 34=I2C 35=I2C 36=I2C 37=I2C 38=I2C 39=I2C 40=I2C 41=I2C 42=I2C 43=I2C 44=I2C 45=I2C 46=I2C 47=I2C 48=I2C 49=I2C 50=I2C 51=I2C 52=I2C 53=I2C 54=I2C 55=I2C 56=I2C 57=I2C 58=I2C 59=I2C 60=I2C 61=I2C 62=I2C 63=I2C 64=I2C 65=I2C 66=I2C 67=I2C 68=I2C 69=I2C 70=I2C 71=I2C 72=I2C 73=I2C 74=I2C 75=I2C 76=I2C 77=I2C 78=I2C 79=I2C 80=I2C 81=I2C 82=I2C 83=I2C 84=I2C 85=I2C 86=I2C 87=I2C 88=I2C 89=I2C 90=I2C 91=I2C 92=I2C 93=I2C 94=I2C 95=I2C 96=I2C 97=I2C 98=I2C 99=I2C 100=I2C 101=I2C 102=I2C 103=I2C 104=I2C 105=I2C 106=I2C 107=I2C 108=I2C 109=I2C 110=I2C 111=I2C 112=I2C 113=I2C 114=I2C 115=I2C 116=I2C 117=I2C 118=I2C 119=I2C 120=I2C 121=I2C 122=I2C 123=I2C 124=I2C 125=I2C 126=I2C 127=I2C 128=I2C 129=I2C 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